

COSC 594: Interplay Between RL and GenAI

Homework 3

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This homework is due on **10/24, 11:59pm**. Write your solution in Latex and submit it on Canvas.

1. (DM alignment)

- (a) A pretrained diffusion model is given by a backward process that begins with x_T sampled from the standard Gaussian $p(x_T) = \mathcal{N}(x_T; 0, I)$, and iteratively transitions from x_T to x_0 through conditional distributions $p_t^\theta(x_{t-1} | x_t)$ for $t = T, \dots, 1$, where θ is the model parameter. The joint distribution of $(x_0, x_1, \dots, x_T)$ is given by

$$p^\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_t^\theta(x_{t-1} | x_t).$$

Recall the diffusion model is represented as a Gaussian with mean $\mu_\theta(x_t, t)$ and variance $\sigma^2(t)$,

$$p_t^\theta(x_{t-1} | x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma^2(t)I).$$

In particular, we consider a class of score-based diffusion models,

$$\mu_\theta(x_t, t) = \frac{1}{\alpha_t} x_t + \frac{1 - \alpha_t}{\sqrt{\alpha_t}} s_\theta(x_t, t)$$

where $s_\theta(x_t, t)$ is the score model that approximates the true score function of a forward process. Recall the KL divergence of distribution q from distribution p by $D_{\text{KL}}(q \| p) := \mathbb{E}_{x \sim q(x)} \log(\frac{q(x)}{p(x)})$.

- i. (1pt) Given a pretrained diffusion model with parameter θ^{pre} , compute the following path-wise KL divergences between this pretrained

model $p^{\theta^{\text{pre}}}(x_{0:T})$ and a finetuned model $p^\theta(x_{0:T})$ in terms of their score models: $s_\theta(x_t, t)$, $s_{\theta^{\text{pre}}}(x_t, t)$,

$$D_{\text{KL}}(p^\theta(x_{0:T}) \| p^{\theta^{\text{pre}}}(x_{0:T})) \text{ and } D_{\text{KL}}(p^{\theta^{\text{pre}}}(x_{0:T}) \| p^\theta(x_{0:T})).$$

Solution:

- (b) Instead of using the reverse KL divergence as we did in class, we consider the following finetuning problem with forward KL divergence,

$$\underset{p}{\text{maximize}} \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \beta \mathbb{E}_{x_{0:T} \sim p^{\theta^{\text{pre}}}(\cdot)} \left[\sum_{t=T+1}^1 D_{\text{KL}}(p_t^{\theta^{\text{pre}}}(\cdot | x_t) \| p_t(\cdot | x_t)) \right]$$

where we denote the optimal solution by $p^*(x_{0:T})$. Define the optimal value function at t as

$$\begin{aligned} v_{t-1}(x_{t-1}) &= \mathbb{E}_{x_{t-2} \sim p_{t-1}^*(\cdot | x_{t-1}), \dots, x_0 \sim p_1^*(\cdot | x_1)} \left[r(x_0) \right] \\ &\quad - \beta \mathbb{E}_{x_{t-2} \sim p_{t-1}^{\theta^{\text{pre}}}(\cdot | x_{t-1}), \dots, x_0 \sim p_1^{\theta^{\text{pre}}}(\cdot | x_1)} \left[\sum_{k=t-1}^1 D_{\text{KL}}(p_k^{\theta^{\text{pre}}}(\cdot | x_k) \| p_k^*(\cdot | x_k)) \right]. \end{aligned}$$

By default, $v_0(x_0) = r(x_0)$.

- i. (1pt) Use the dynamic programming principle to express the optimal solution p^* in terms of v_t .

Solution:

- ii. (1pts) Find the recursive expression of v_t i.e., express v_t in terms of v_{t-1} , and use this recursion to express v_t in terms of $r(x_0)$.

Solution:

- iii. (2pts) Use the importance sampling to design an online training-free sampling algorithm that allows you to sample from $p^*(x_0)$. *Hint:* formulate a regression problem to estimate v_t and apply self-normalized importance sampling to $p^*(x_{0:T})$ with the estimated v_t .

Solution:

2. (DM composition)

- (a) Given K pretrained models $\{p_k^{\theta^{\text{pre}}}\}_{k=1}^K$, each is in the form of the score model $s_{\theta_k^{\text{pre}}}(x_t, t)$. We use shorthand p^k for $p_k^{\theta^{\text{pre}}}$, and $s_k(x_t, t)$ for

$s_{\theta_k^{\text{pre}}}(x_t, t)$. We consider the following finetuning problem with point-wise forward KL divergence,

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(p^k(\cdot) \| p(\cdot)).$$

- i. (0.5pt) Find a closed-form expression of the optimal policy p^* .

Solution:

- ii. (0.5pt) Relax the point-wise KL divergence to the path-wise one, and replace p by the score-based diffusion model (refer to 1(a)). Compute the policy gradient of the objective function on θ .

Solution:

- (b) We consider the following finetuning problem with point-wise forward KL divergence constraints,

$$\begin{aligned} &\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] \\ &\text{subject to} \quad D_{\text{KL}}(p^k(\cdot) \| p(\cdot)) \leq \epsilon_k \text{ for } k = 1, \dots, K. \end{aligned}$$

- i. (1pt) Find a closed-form expression of the dual function $D(\lambda)$,

$$D(\lambda) = \underset{p}{\text{maximize}} \quad L(p, \lambda)$$

where $L(p, \lambda) = \mathbb{E}_{x_0 \sim p(\cdot)}[r(x_0)] + \sum_{k=1}^K \lambda_k (\epsilon_k - D_{\text{KL}}(p^k(\cdot) \| p(\cdot)))$ is the Lagrangian with dual variables $\lambda_k \geq 0$ for $k = 1, \dots, K$.

Solution:

- ii. (1pt) Check the convexity of the dual function $D(\lambda)$, and find an optimal dual variable λ^* ,

$$\lambda^* \in \underset{\lambda \geq 0}{\text{argmax}} \quad D(\lambda).$$

Solution:

- iii. (2pts) Assume the strong duality holds. This allows us to find an optimal solution p^* via

$$p^* \in \underset{p}{\text{argmax}} \quad L(p, \lambda^*).$$

Compute one such optimal solution. Design a training algorithm that allows us to approximately find this solution. *Hint:* relax the

point-wise KL divergence to the path-wise one, and apply dual or primal-dual algorithms.

Solution: